UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/570,822	01/07/2022	Yu Fang	SONA-0007-U01-C03	3577

87084 7590 02/16/2023

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
ALGIBHAH, HAMZA N	

ART UNIT	PAPER NUMBER
2456	

NOTIFICATION DATE	DELIVERY MODE
02/16/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/570,822	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2456	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01/07/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/07/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-20*** are pending.

2. ***Claims 1-20 are rejected.***

Double Patenting

3. A rejection based on double patenting of the "same invention" type finds its support in the language of 35 U.S.C. 101 which states that "whoever invents or discovers any new and useful process... may obtain a patent therefor..." (Emphasis added). Thus, the term "same invention," in this context, means an invention drawn to identical subject matter. See *Miller v. Eagle Mfg. Co.*, 151 U.S. 186 (1894); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Ockert*, 245 F.2d 467, 114 USPQ 330 (CCPA 1957).

A statutory type (35 U.S.C. 101) double patenting rejection can be overcome by canceling or amending the claims that are directed to the same invention so they are no longer coextensive in scope. The filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. 101.

4. Claim **2** is rejected under 35 U.S.C. 101 as claiming the same invention as that of claim **11** of Patent No. 11,252,039.

5. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious

over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

6. Claims 1, 3-20 are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1-29 of U.S. Patent No. 11,252,039. Although the claims at issue are not identical, they are not patentably distinct from each other because instant claims are broader in scope comparing to the patent claims.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

7. **Claims 14 and 16** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

8. Claim 14 recites the limitation "the local DNS" in line 2 of the claim. There is insufficient antecedent basis for this limitation in the claim.

9. Claim 16 recites the limitation "the local DNS" in line 3 of the claim. There is insufficient antecedent basis for this limitation in the claim.

Claim Interpretation

The following is a quotation of 35 U.S.C. 112(f):

(f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim

> shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

10. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth

paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are: "a policy manager circuit structured to interpret, a configuration circuit structured to configure, and a gatekeeper interface circuit structured to regulate" in claim 1. The corresponding structures and algorithms used to link to the above limitations are the structures and algorithms stated in the specification (for example paragraph 0101, 0152, 0291-0296, 0307-0310)

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim

limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

11. ***Claims 1 and 13 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by BARRETT et al (Pub. No.: US 2019/0394089 A1).

As per claim 1, BARRETT discloses a system, comprising:

- ***a vehicle having a first network zone and a second network zone of a different type than the first network zone*** (BARRETT, Fig 1, 2, paragraph 0021, 0027-0028 vehicle network, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay)***;***
- ***a gatekeeper interface circuit interposed between the first network zone and a transceiver selectively couplable to an external device*** (BARRETT, Fig 2, paragraph 0037, wherein for example the gateway 240 can be the gatekeeper interface circuit which is between internal network nodes of a first type (such as CAN) and telecoms control unit 260 (the transceiver) that is connected to an external network)***, and further interposed between the second network zone and the transceiver*** (BARRETT, Fig 2, paragraph 0037, wherein for

example the gateway 240 can be the gatekeeper interface circuit which is between internal network nodes of a second type (such as LIN and/or FlexRay) and telecoms control unit 260 (the transceiver));

- a policy manager circuit structured to interpret a policy comprising an external data routing description (BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action. The firewall system can provide inter-domain firewalling, intra-domain firewalling, or both); and

- a configuration circuit structured to configure the gatekeeper interface circuit in response to the external data routing description (BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system of the vehicle network is configured based on the connectivity information. In some implementations, configuring a firewall system of the vehicle network includes configuring one or more firewalls in the vehicle network based on the connectivity information (e.g., the connectivity information recorded in Table 1 or 2). In some implementations, configuring one or more firewalls in the vehicle network based on the connectivity information includes configuring a firewall (e.g., on a gateway node, a DC, or another node) in the vehicle network to secure intra-domain communications, inter-domain communications, or both)

wherein

- ***the gatekeeper interface circuit is structured to regulate communications between end***

points of the first network zone and the transceiver, and to regulate communications between end points of the second network zone and the transceiver (BARRETT, Fig 2, paragraph 0037, inter-domain firewalling can ensure that only certain types of traffic can pass from the external network 205 into the trusted network 220 internal to the vehicle 202. An internal firewall, FW.sub.int, 230 is placed between the switch or router 210 of the the trusted network 220 and the gateway 240 of the IZ 225. An external firewall, FW.sub.ext, 250 is placed between the gateway 240 of the IZ 225 and the telecoms control unit 260);

Claim 13 is rejected under the same rationale as claim 1.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

12. **Claims 14, 16-20 are** rejected under 35 U.S.C. 103 as being unpatentable over BARRETT et al (Pub. No.: US 2019/0394089 A1) in view of Lei et al (Pub. No.: US 2017/0339095 A).

As per claim 14, claim 13 is incorporated and BARRETT does not explicitly disclose determining an address change of an end point of the first network zone or second network zone, and updating the local DNS in response to the address change. However, Lei discloses ***determining an address change of an end point of the first network zone or second network zone, and updating the local DNS in response to the address change*** (Lei, Fig 1, 4-5, paragraph

0059, 0027-0028, wherein the vehicle 102 sends the updated network address 120 to the domain name server 142 to update the DNS records 144. In an example, the telematics control unit 116 may automatically send a DNS record change message over the communications network 114 to the domain name server 142. The DNS record change message may indicate the new network address 120 of the vehicle 102 and the hostname 146 of the vehicle 102. Responsive to receipt by the domain name server 142, the domain name server 142 may update the network address 120 associated with the hostname 146 of the vehicle 102).

Therefore, it would have been obvious to one ordinary skill in the art before the effective the filing date of the invention to modify BARRETT using Lei to update DNS as claimed because this would have provided way to ensure the stability of the network connection of the system and thus improving the efficiency and performance of the system.

As per claim 16, claim 13 is incorporated and BARRETT does not explicitly disclose determining a change of a non-local address value for an end point of the first network zone or second network zone, and updating the local DNS in response to the non-local address value change. However, Lei discloses ***determining a change of a non-local address value for an end point of the first network zone or second network zone, and updating the local DNS in response to the non-local address value change*** (Lei, Fig 1, 4-5, paragraph 0059, 0027-0028, wherein the vehicle 102 sends the updated network address 120 to the domain name server 142 to update the DNS records 144. In an example, the telematics control unit 116 may automatically send a DNS record change message over the communications network 114 to the domain name server 142. The DNS record change message may indicate the new network address 120 of the vehicle 102 and the hostname 146 of the vehicle 102. Responsive to receipt

by the domain name server 142, the domain name server 142 may update the network address 120 associated with the hostname 146 of the vehicle 102).

Therefore, it would have been obvious to one ordinary skill in the art before the effective the filing date of the invention to modify BARRETT using Lei to update DNS as claimed because this would have provided way to ensure the stability of the network connection of the system and thus improving the efficiency and performance of the system.

As per claim 17, claim 16 is incorporated and Lie discloses determining an authorization of an external device providing the change of the non-local address value (Lei, Fig 1, 4-5, paragraph 0034, wherein the firewall 126 may implement address filtering to prevent commands from being requested by the web control application 138 of an illegitimate mobile device 122. The firewall 126 may accordingly implement validation functionality, such as mobile device 122 certificate verification, to ensure that connected mobile devices 122 are authorized to use the telematics control unit 116 functionality of the vehicle 102).

As per claim 18, claim 13 is incorporated and BARRETT does not explicitly disclose wherein the external data routing description comprises an external domain name server (DNS) comprising a plurality of external address values for external network access locations, each corresponding to at least one local communicating device. However, Lei discloses ***wherein the external data routing description comprises an external domain name server (DNS) comprising a plurality of external address values for external network access locations, each corresponding to at least one local communicating device*** (Lei, Fig 1, 4-5, paragraph 0024, wherein A virtual DNS server application installed to the user's mobile device may act as a

virtual DNS server, and may monitor and maintains IP address mapping overrides for vehicle-specific hostname URLs (e.g., for www.Dan-ford-F150.ford.com in this example)).

Therefore, it would have been obvious to one ordinary skill in the art before the effective the filing date of the invention to modify BARRETT using Lei to update DNS as claimed because this would have provided way to ensure the stability of the network connection of the system and thus improving the efficiency and performance of the system.

As per claim 19, claim 18 is incorporated and BARRETT discloses wherein each at least one local communicating device comprises at least one of: an end point of the first network zone or the second network zone; an application; a flow; a vehicle function; or a vehicle controller (BARRETT, Fig 2-3, paragraph 0037, wherein the trusted network 220 is connected to the IZ 225 via a switch or router 210, which is further connected to a gateway 240 of the IZ 225 for connection to the external network 205 through the telecoms control unit 260. In one example implementation, an internal firewall, FW.sub.int, 230 is placed between the switch or router 210 of the the trusted network 220 and the gateway 240 of the IZ 225. An external firewall, FW.sub.ext, 250 is placed between the gateway 240 of the IZ 225 and the telecoms control unit 260).

As per claim 20, claim 18 is incorporated and BARRETT discloses wherein each external network access location comprises at least one of: an internet address; a wide area network address; an external device identifier; or an external application identifie (BARRETT, Fig 2-3, paragraph 0060, wherein in the example case of IP based connectivity, the IP address, protocol type, and port number of the listening services can be recorded. In some implementations, a dynamic analysis includes turning off all firewalls and exercising the various software states of

the vehicle using a packet sniffing application to record connectivity used (source and destination addresses, etc.)).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached on 7:30 am - 3:30 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Umar Cheema can be reached on (571) 270-3037. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2456